

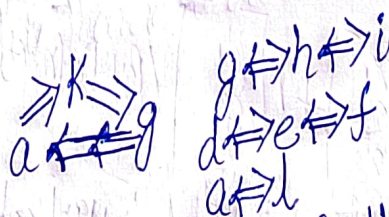
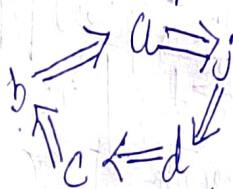
2.3 Characterizations of Invertible Matrices

Theorem 8: The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.

- A is an invertible matrix.
- A is row equivalent to the $n \times n$ identity matrix.
- A has n pivot positions.
- The equation $Ax=0$ has only the trivial solution.
- The columns of A form a linearly independent set.
- The linear transformation $x \mapsto Ax$ is one-to-one.
- The equation $Ax=b$ has at least one solution for each b in $\mathbb{R}^n$.
- The columns of A span $\mathbb{R}^n$.
- The linear transformation $x \mapsto Ax$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- There is an $n \times n$ matrix C such that $CA=I$.
- There is an $n \times n$ matrix D such that $AD=I$.
- A^T is an invertible matrix.

We shall establish the following implications in order to show that the statements are equivalent.



Proof: ~~$a \Rightarrow j$~~ Let, A be an invertible matrix. By the defn. of inverse, there exists a unique matrix of size $n \times n$, A^{-1} s.t. $A^{-1}A=I$.
 $\therefore C=A^{-1}$. $\therefore a \Rightarrow j$ holds.

$j \Rightarrow d$ This is True by Prob 23 from Ex-2.1.
 $d \Rightarrow c$ The equation $Ax=0$ has only the trivial solution. By Theorem 3 from Chapter 1, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$. Since the solution is unique, By Theorem 2 from